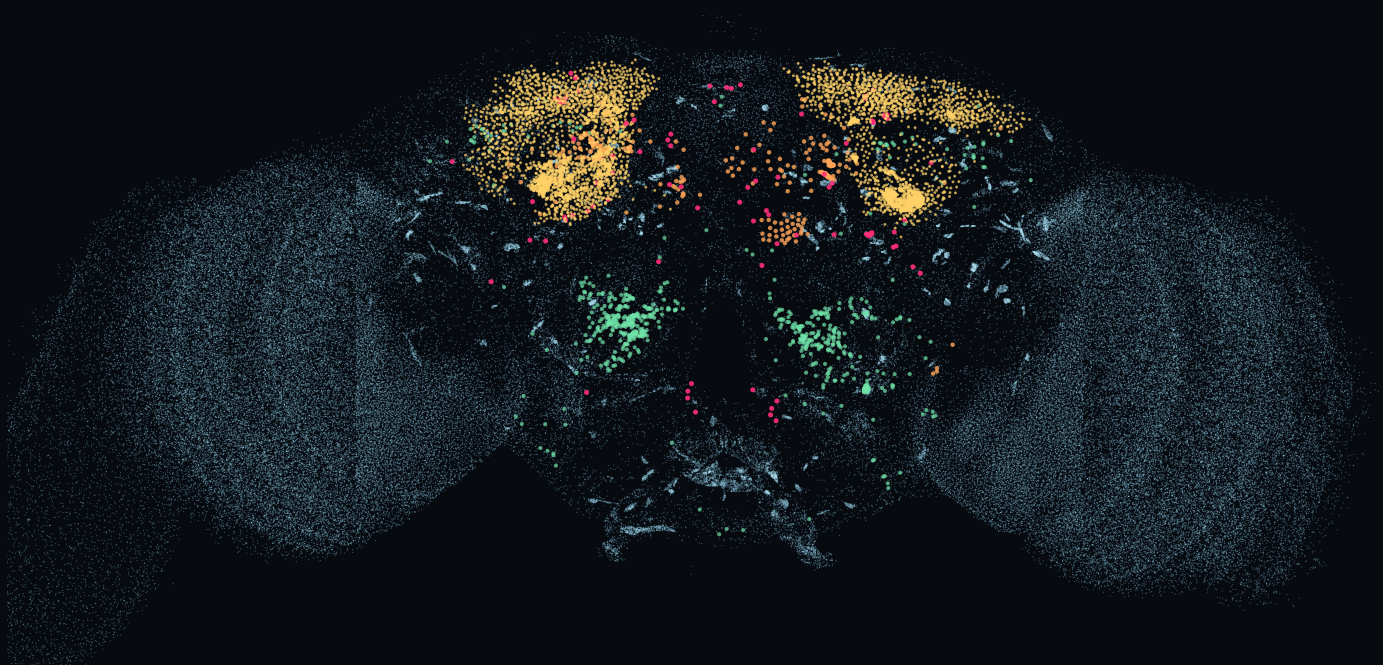


FlyScale

Geometric scaling of the adult *Drosophila* connectome

139,255 neurons · 2,700,513 connections · 34,153,566 synapses · a 10x ladder

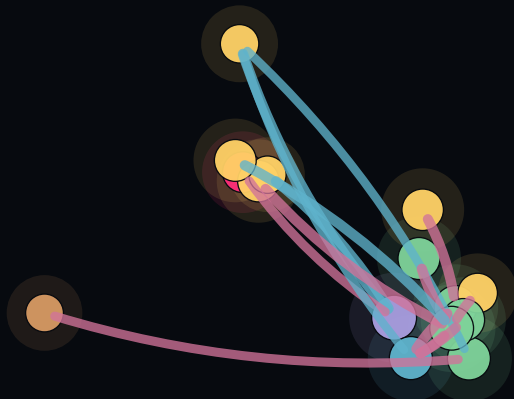
to answer whether a connectome can be geometrically enlarged while preserving its structure, dynamics, capabilities and measured energy budget



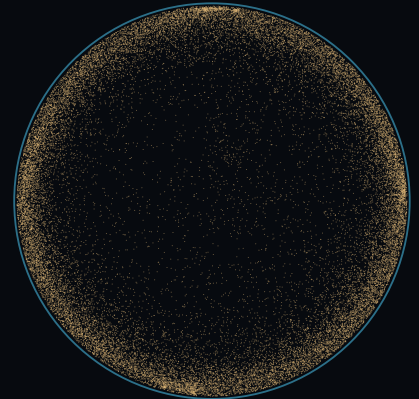
The whole brain — 139,255 neurons at their soma positions

Gold Kenyon cells · magenta MBON · orange DAN · green antennal-lobe projection neurons

Detail: a real 15-neuron circuit (14 connections, 14303 synapses)



The fitted 2-D hyperbolic geometry



Every mark on this cover is measured data. The brain is the FlyWire v783 connectome, all 139,255 neurons at their soma positions, with the mushroom-body circuit highlighted. The small panel is a real circuit (edge width = synapse count, edge color = predicted transmitter, positions rescaled to fit). The disk is the hyperbolic geometry fitted to the same connectome in section 5.